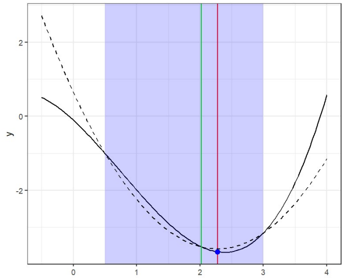
Optimization in Machine Learning

Univariate optimization
Brent's method



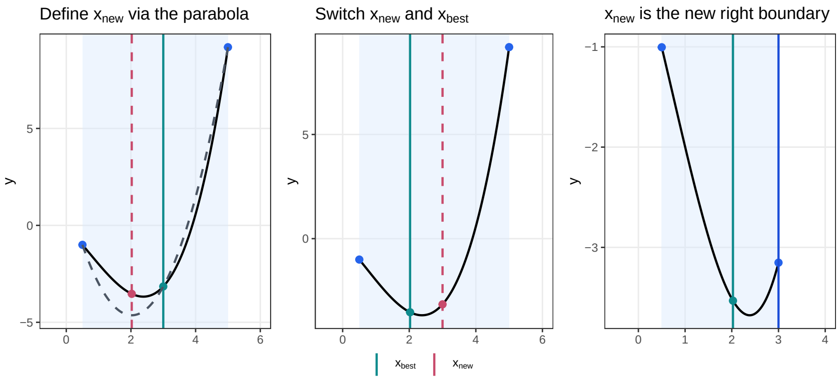
Learning goals

- Quadratic interpolation
- Brent's procedure

Slides based on ▸ Press et al. 1992 (Ch. 10.2)

QUADRATIC INTERPOLATION

Similar to golden section search but select x_{new} differently:
minimum of parabola through $(x_{\text{left}}, f_{\text{left}}), (x_{\text{best}}, f_{\text{best}}), (x_{\text{right}}, f_{\text{right}})$



Left: Fit parabola (dashed) and propose its minimum (red) as new point.

Middle: Switch x_{new} and x_{best} if $f(x_{\text{new}}) < f_{\text{best}}$. Right: New interval.

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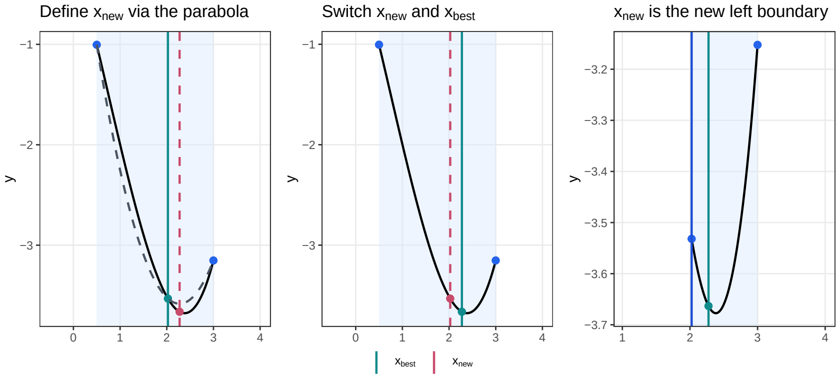

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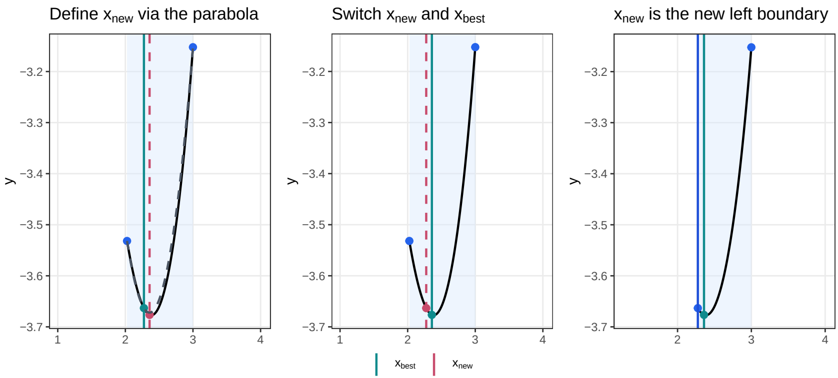

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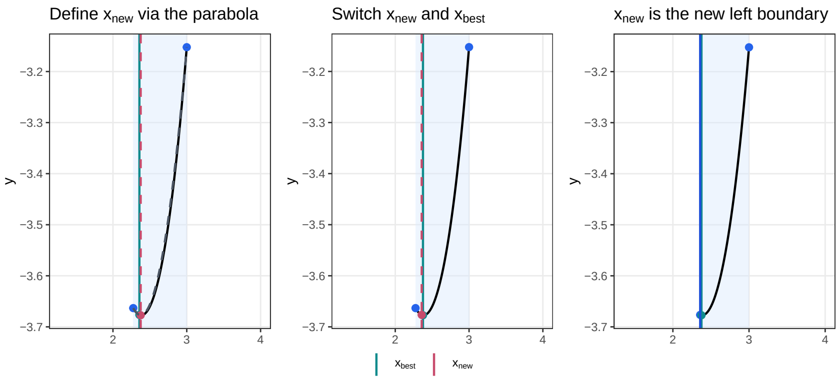

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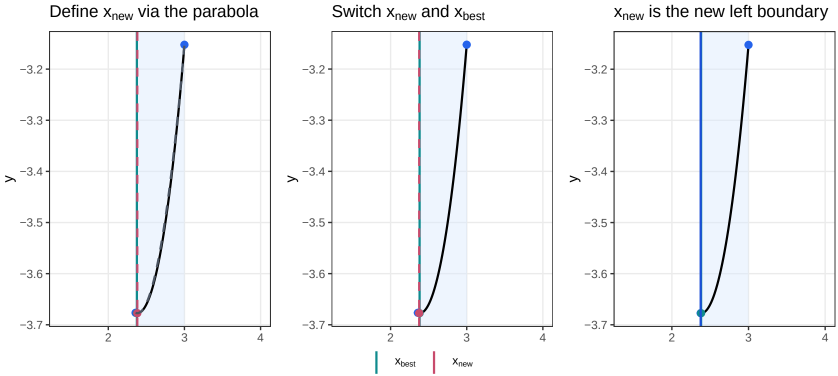

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QUADRATIC INTERPOLATION COMMENTS

- Much faster than GSS for smooth f: well-approximated by a parabola near the minimum $\Rightarrow$ QI jumps directly toward the minimum
- But: not as robust as GSS, the following may happen:
  - Algorithm suggests very similar x_{new} in each step
  - x_{new} outside of search interval
  - Parabola degenerates to line and no real minimum exists
- Algorithm must then abort, finding the minimum is not guaranteed

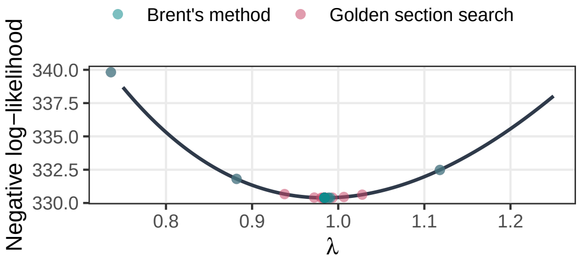
BRENT'S METHOD

- Brent's algorithm ▸ Brent 1973 alternates between golden section search and quadratic interpolation as follows:
  - Quadratic interpolation step acceptable if:
    (i) x_{new} falls within $[x_{\text{left}}, x_{\text{right}}]$
    (ii) x_{new} sufficiently far away from x_{best}
    (Heuristic: Less than half of the movement of step before last)
  - Otherwise: proposal via GSS
- GSS stabilizes unstable steps and guarantees **at least linear convergence** for any unimodal f
- Faster convergence (quadratic interpolation) on well-behaved (smooth) f: **superlinear convergence** of order ≈ 1.325
- Note: Brent's algorithm also exists with other robust methods than GSS, e.g., bisection method

EXAMPLE: MLE POISSON

- Poisson PMF: $p(k|\lambda) := \mathbb{P}(x = k) = \frac{\lambda^k \cdot \exp(-\lambda)}{k!}$
- Negative log-likelihood for n samples:

$$-\ell(\lambda, \mathcal{D}) = -\log \prod_{i=1}^{n} p(x^{(i)}|\lambda) = -\sum_{i=1}^{n} \log p(x^{(i)}|\lambda)$$



GSS and Brent converge to global min at $x^* \approx 1$

But GSS needs ≈ 45 iters, Brent only ≈ 15 for same tolerance